

Dongfang Zhao, Ph.D.

Nationality: United States of America

Education

University of Washington (UW), Seattle	Computer Science, PostDoc	2017
Illinois Institute of Technology, Chicago	Computer Science, PhD	2015
Emory University, Atlanta, GA	Computer Science, MS	2008
Katholieke Universiteit Leuven, Belgium	Statistics, Diploma	2005

Employment

2025–now	Adjunct Assistant Professor, UW Paul G. Allen School of Computer Science & Engineering
2023–now	Tenure-Track Assistant Professor, UW Tacoma School of Engineering & Technology
2018–2023	Affiliate Faculty, University of California, Davis
2018–2023	Assistant Professor, University of Nevada, Reno
2015	Research Intern, IBM Almaden Research Center, San Jose, CA
2014	Research Intern, Argonne National Laboratory, Chicago, IL
2013	Research Intern, Pacific Northwest National Laboratory, Seattle, WA
2009–2011	Software Developer (full-time), Epic Systems Corporation, Madison, WI

Awards

2022	Federal Government Research Grant, NSF 2112345, \$255,916 “A DLT Machine Learning Platform for Blockchain Warehousing”
2020	Federal Government Research Grant, DOE SC0020455, \$200,000 “HPChain: Energy-Efficient Scalable Blockchains for HPC Systems”
2019	Best Paper Award, International Conference on Cloud Computing “BlockLite: Toward Accurate and Efficient Emulation of Public Blockchains in the Cloud”
2018	Best Student Paper Award, IEEE International Conference on Cloud Computing “Toward Cost-Effective Memory Scaling in Clouds: Symbiosis of Virtual and Physical Memory”

Recent Projects

[BadRSSD]	Information-theoretical backdoor attacks on diffusion models utilizing residual score subspace dispersive loss: https://github.com/hpdc/BadRSSD (private for now) <i>Tags:</i> Generative AI security, Latent subspace learning, Fine-tuning integrity, PyTorch
[ElasticIVF]	Elastic in-memory vector indexing and search for streaming data analysis on heterogeneous distributed platforms: https://github.com/hpdc/ElasticIVF <i>Tags:</i> Distributed GPU computing, Message passing interface (MPI), Faiss, CUDA kernel
[AdaDisk]	Decentralized agentic systems for adaptive ingestion-query scheduling of disk-resident vector indices in LLM serving: https://github.com/hpdc/AdaDisk <i>Tags:</i> Large language models, AI agents, Distributed systems, DiskANN, C++, Python
[Hermes]	MySQL plug-in for high-performance homomorphic encryption of vector databases in privacy-preserving RAG: https://github.com/hpdc/Hermes <i>Tags:</i> Retrieval-augmented generation, Fully homomorphic encryption, OpenFHE, C++

Selected Publications (Dongfang Zhao was/is the *student's academic advisor)

- [1] *Baigang Chen, **Dongfang Zhao**. Reliable Non-Leveled Homomorphic Encryption for Web Services. The ACM Web Conference (formerly International World Wide Web Conference, WWW), 2026.
- [2] *Chengyu Gong, *Gefei Shen, Luanzheng Guo, Nathan Tallent, **Dongfang Zhao**. Order-Preserving Dimension Reduction for Multimodal Semantic Embedding. The 40th Annual AAAI Conference on Artificial Intelligence (AAAI), 2026.
- [3] *Jiuzhou Fu, Luanzheng Guo, Nathan Tallent, **Dongfang Zhao**. ProHD: Projection-Based Hausdorff Distance Approximation. IEEE International Conference on Data Mining (ICDM), 2025.
- [4] *Jiayao Wang, . . . , **Dongfang Zhao**. IPBA: Imperceptible Perturbation Backdoor Attack in Federated Self-Supervised Learning. European Conference on Artificial Intelligence (ECAI), 2025.
- [5] *Olamide Timothy Tawose, Jun Dai, Lei Yang, **Dongfang Zhao**. Toward Efficient Homomorphic Encryption for Outsourced Databases through Parallel Caching. Proceedings of the ACM on Management of Data (SIGMOD), 2023.
- [6] *Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**. SciChain: Blockchain-enabled Lightweight and Efficient Data Provenance for Reproducible Scientific Computing. 37th IEEE International Conference on Data Engineering (ICDE), 2021.
- [7] *Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**. BAASH: lightweight, efficient, and reliable blockchain-as-a-service for HPC systems. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC), 2021.
- [8] *Xinying Wang, *Olamide Timothy Tawose, Feng Yan, **Dongfang Zhao**. HDK: Toward High-Performance Deep-Learning-Based Kirchhoff Analysis. The Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI), 2020.
- [9] Heyang Qin, Syed Zawad, Yanqi Zhou, Lei Yang, **Dongfang Zhao**, Feng Yan. Swift machine learning model serving scheduling: a region based reinforcement learning approach. The International Conf. for High Performance Computing, Networking, Storage and Analysis (SC), 2019.
- [10] Parmita Mehta, Sven Dorkenwald, **Dongfang Zhao**, Tomer Kaftan, Alvin Cheung, Magdalena Balazinska, Ariel Rokem, Andrew J. Connolly, Jacob VanderPlas, Yusra AlSayyad. Comparative Evaluation of Big-Data Systems on Scientific Image Analytics Workloads. Proceedings of Very Large Data Bases (VLDB), 2017.
- [11] **Dongfang Zhao**, Kan Qiao, Jian Yin, Ioan Raicu. Dynamic Virtual Chunks: On Supporting Efficient Accesses to Compressed Scientific Data. IEEE Transactions on Services Computing (TSC). 9(1): 96-109 (2016).
- [12] **Dongfang Zhao**, Ning Liu, Dries Kimpe, Robert B. Ross, Xian-He Sun, Ioan Raicu. Towards Exploring Data-Intensive Scientific Applications at Extreme Scales through Systems and Simulations. IEEE Transactions on Parallel and Distributed Systems (TPDS). 27(6): 1824-1837 (2016).
- [13] **Dongfang Zhao**, Li Yang. Incremental Isometric Embedding of High-Dimensional Data Using Connected Neighborhood Graphs. IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI). 31(1): 86-98 (2009).